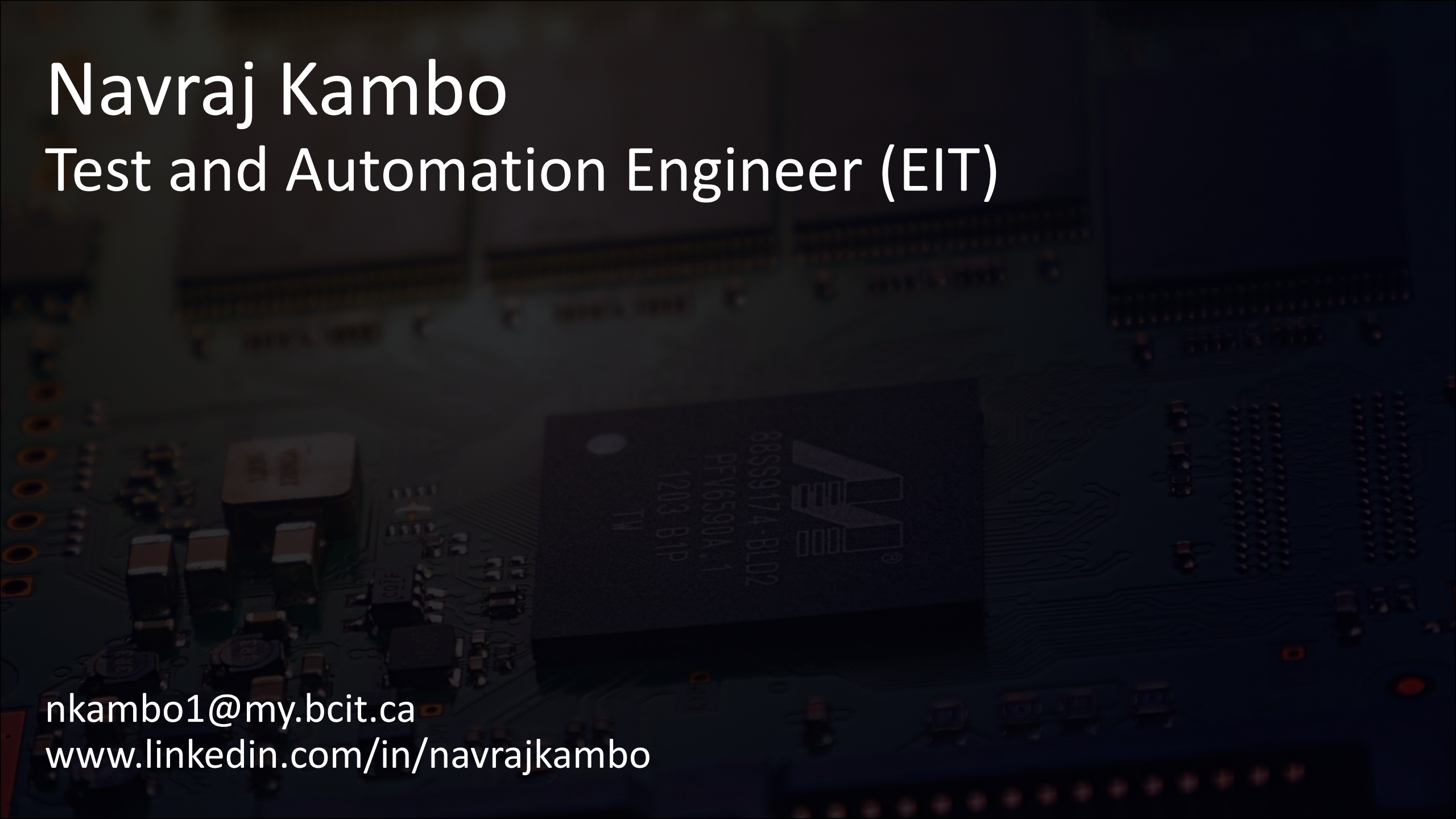




Navraj Kambo

Test and Automation Engineer (EIT)

nkambo1@my.bcit.ca

www.linkedin.com/in/navrajkambo

Navraj Kambo | Overview

- 3 About Me
- 4 Digital Speech Decoding (WTS Wireless Technical Services Inc. / FortisBC)
- 5 NDART (British Columbia Institute of Technology)
- 6 Autonomous Vineyard Harvesting (British Columbia Institute of Technology SoE)
- 7 Apex Motion Control
- 8 Packaging Line(Apex Motion Control)
- 9 DecoBot (Apex Motion Control)
- 10 Wellons Canada
- 11 Continuous Drying System (Wellons Canada)
- 12 Delta Intelligent Building Technologies (Canada) Inc.
- 13 RED 5 Edge (China) (DIBT)
- 14 Test & Automation (DIBT)



Navraj Kambo | About Me

I'm an electrical engineer who graduated in 2019. I have experience in robotics, automation, and electronics. I enjoy collaborating with multi-disciplinary teams. Outside of work I enjoy playing sports.

Education



BEng Electrical Engineering
Research Intern (4 months)

Experience



4-month co-op



Delta Intelligent Building Technologies

Test & Automation Engineer



Design Engineer

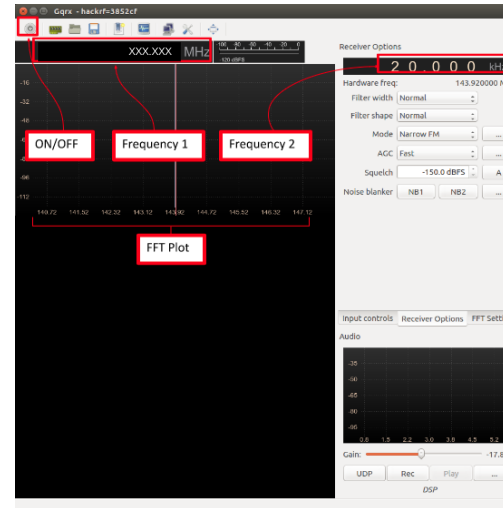


Electrical Engineer EIT

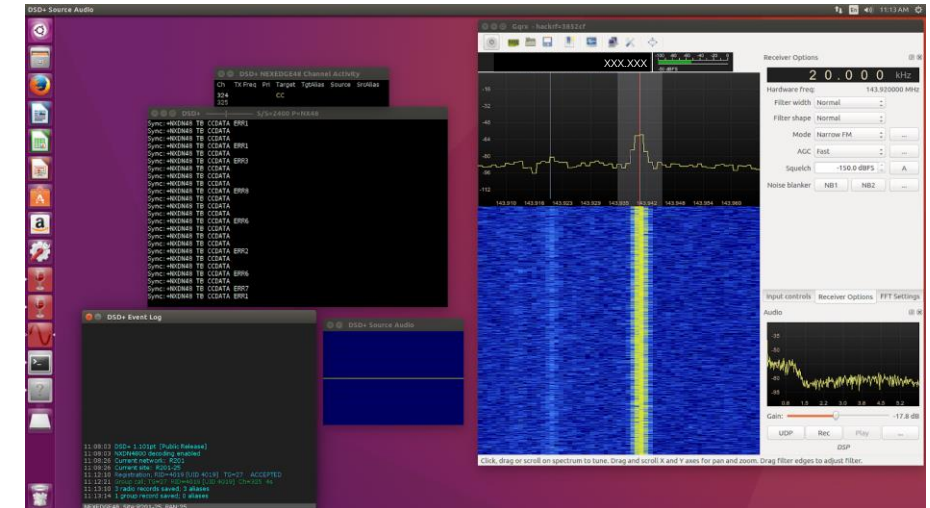
Navraj Kambo | Digital Speech Decoding

```
wts-nav@KRYPTONITE: ~  
wts-nav@KRYPTONITE:~$ hackrf_info  
Found HackRF board 0:  
USB descriptor string: 000000000000000909864c8383852cf  
Board ID Number: 2 (HackRF One)  
Firmware Version: 2015.07.2  
Part ID Number: 0xa000cb3c 0x00594f4f  
Serial Number: 0x00000000 0x00000000 0x909864c8 0x383852cf  
wts-nav@KRYPTONITE:~$
```

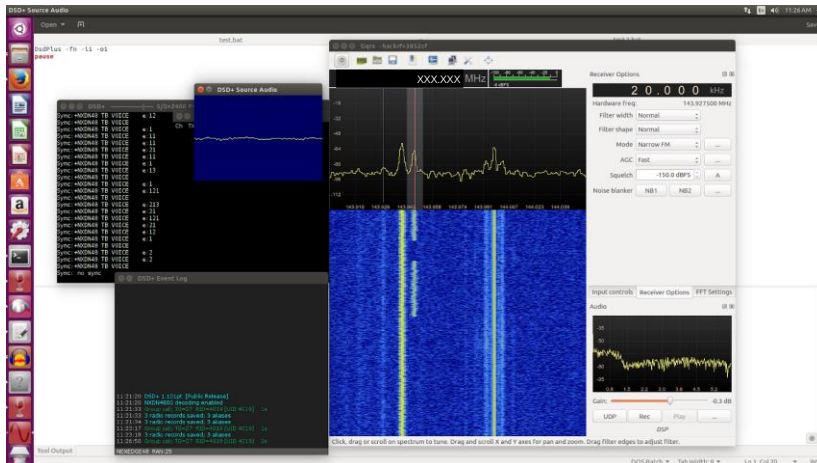
HackRF in Ubuntu



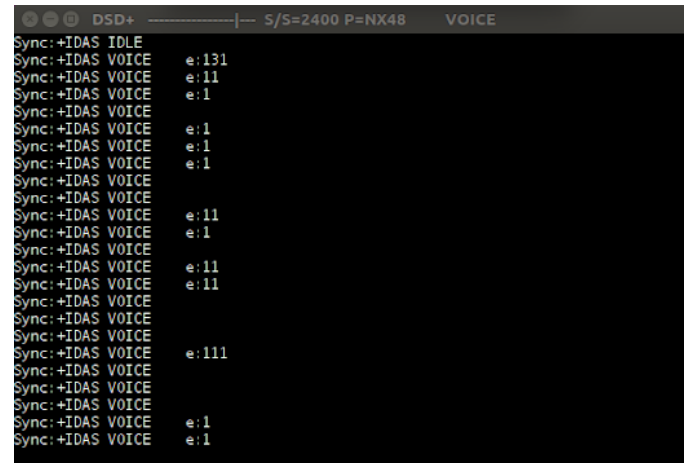
GQRX SDR



Decoding Control Channel (Carrier)



Decoding Traffic Channel (Side-band)

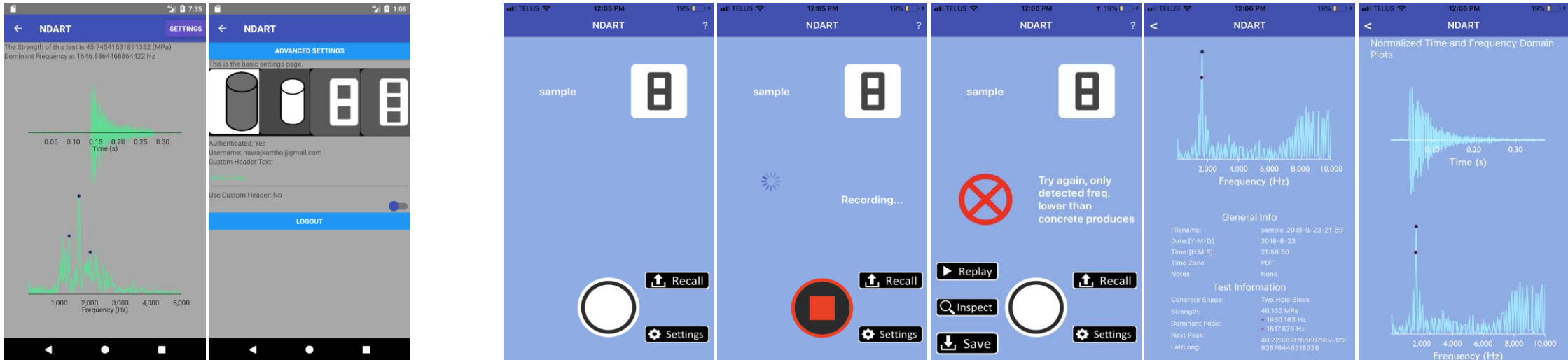


Sync Window on Traffic Channel



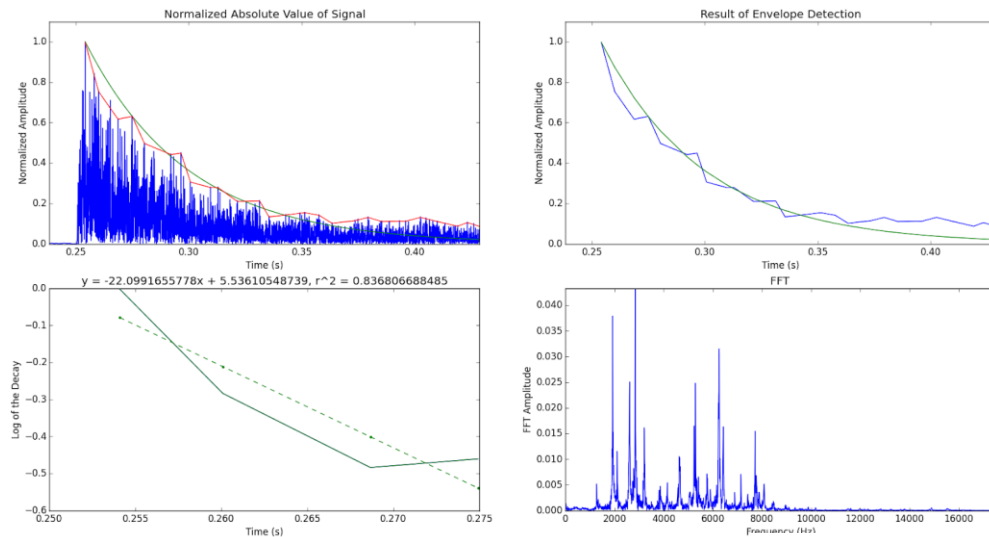
HackRF One

Navraj Kambo | NDART



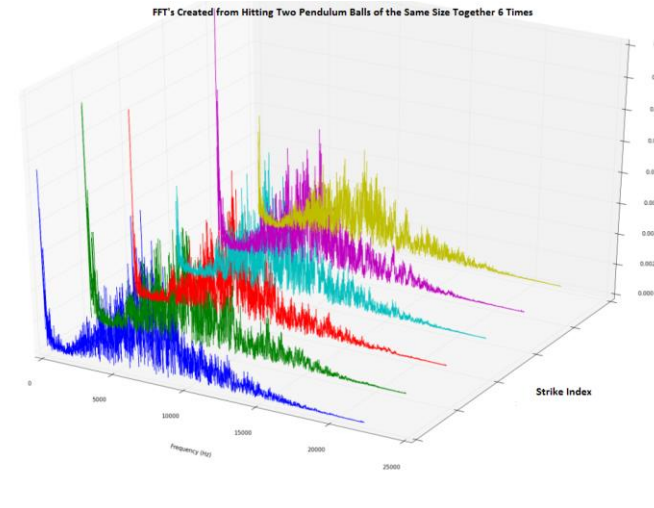
NDART app on Android

NDART app on iOS

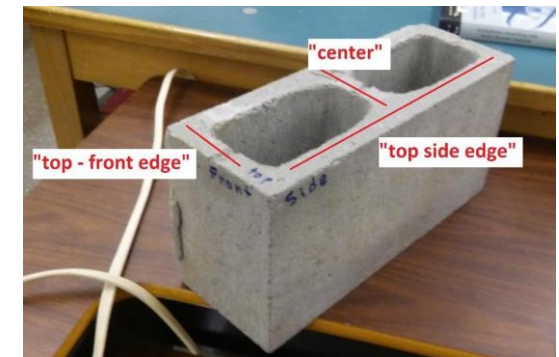


Strike Decay Plot

$$N(t) = N_0 e^{-\lambda t}$$



FFTs at Different Locations

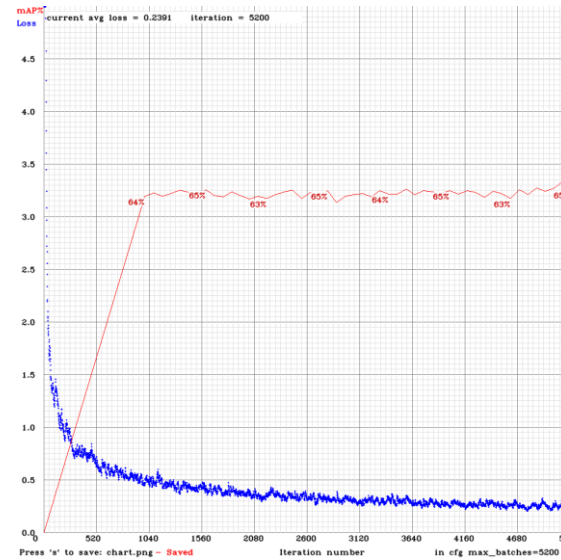


Concrete Block

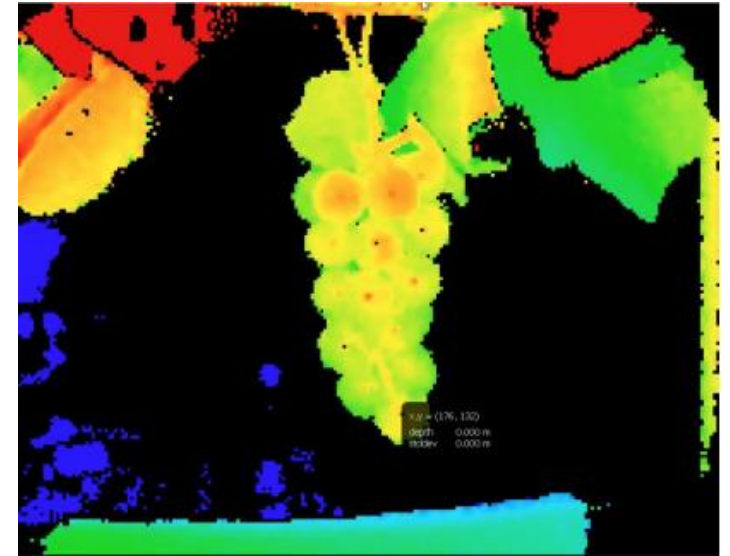
Navraj Kambo | Autonomous Vineyard Harvesting



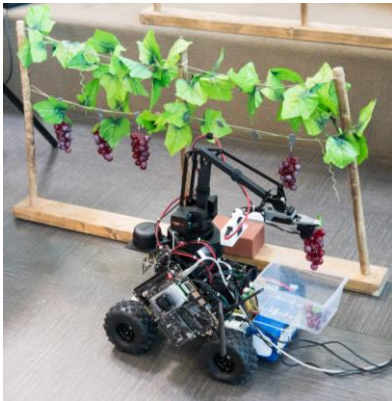
Project Expo



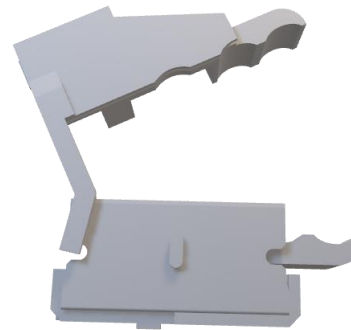
ML Dataset Training



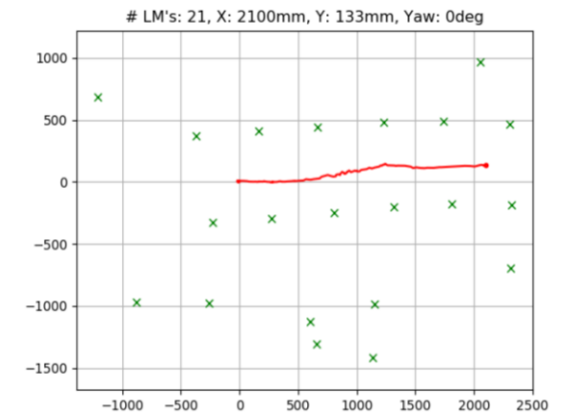
Grape Cluster TOF Image



Vinebot

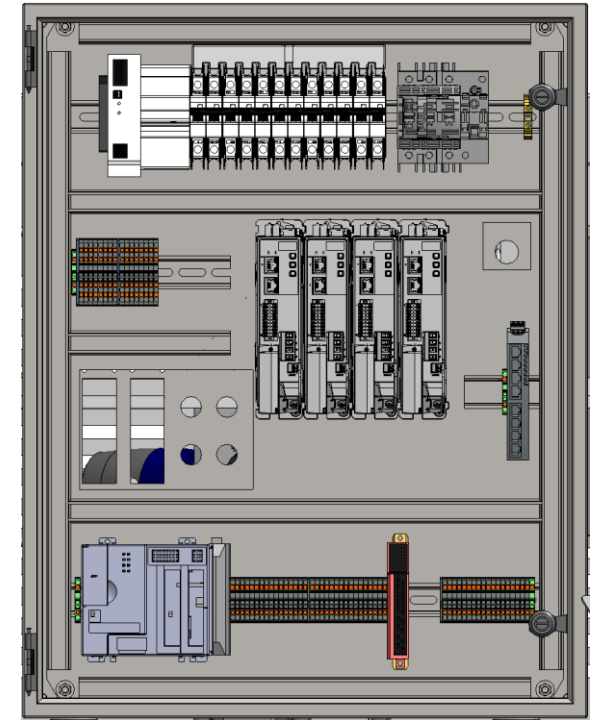
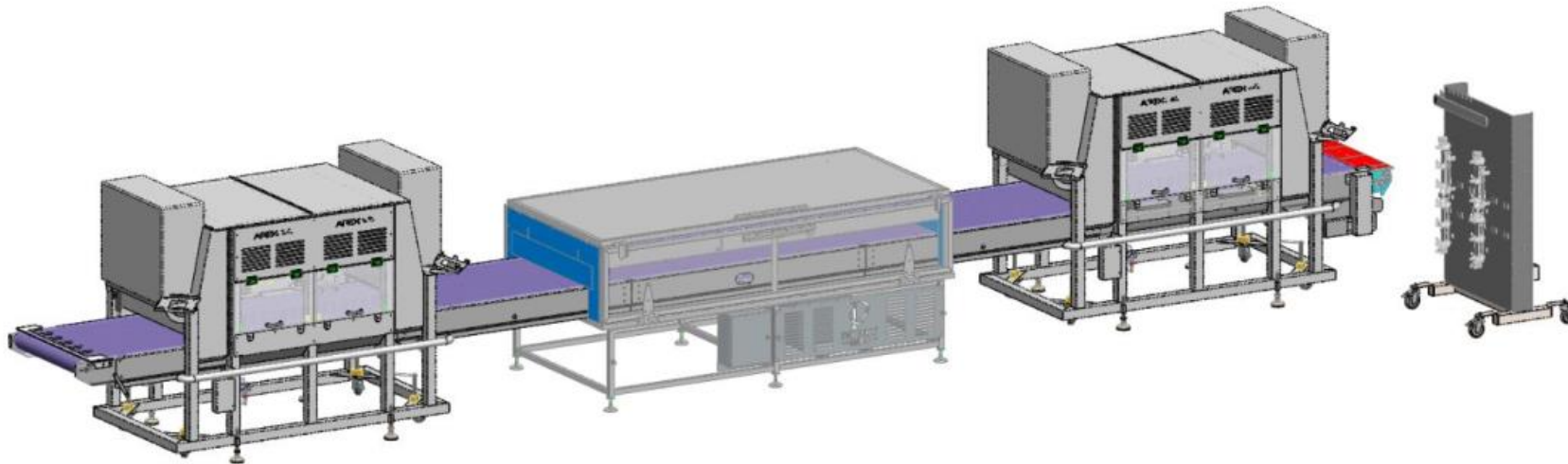


Vinebot Stem Cutter

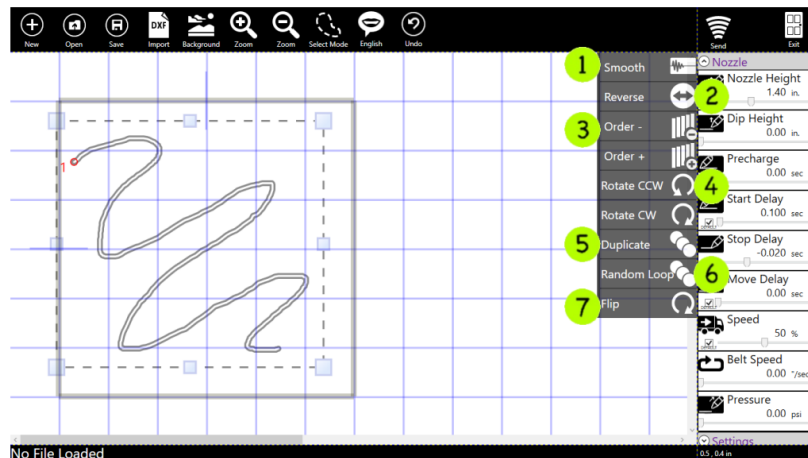


SLAM Path Planning

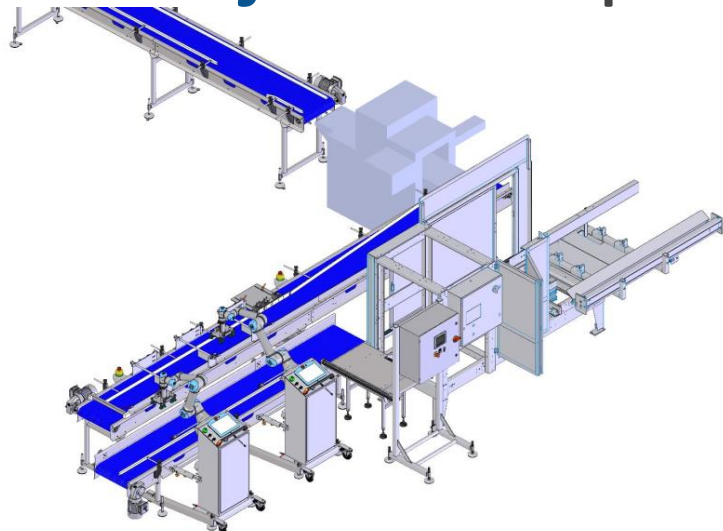
Navraj Kambo | Apex Motion Control - Design Engineer



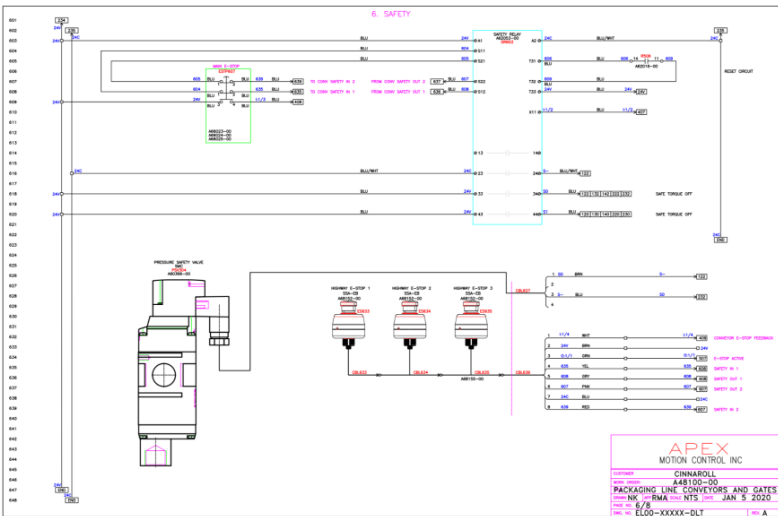
Robotics and Automation (Food and Beverage Industry)



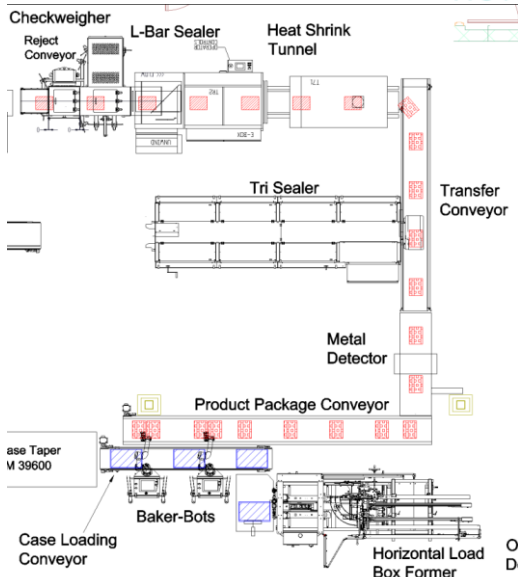
Navraj Kambo | Packaging Line



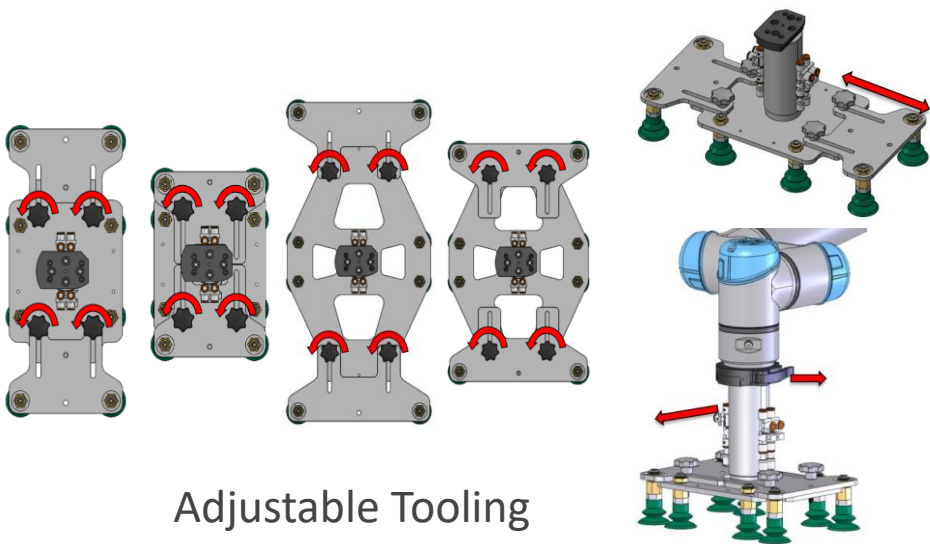
SolidWorks Assembly



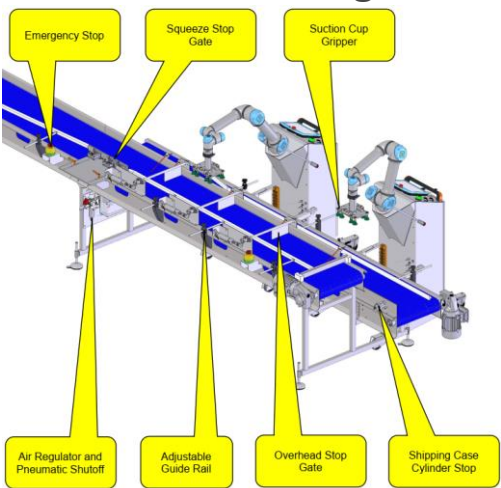
Electrical Design



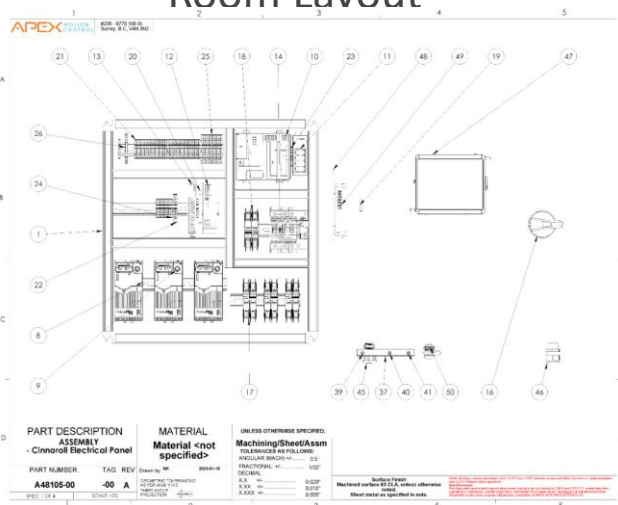
Room Layout



Adjustable Tooling

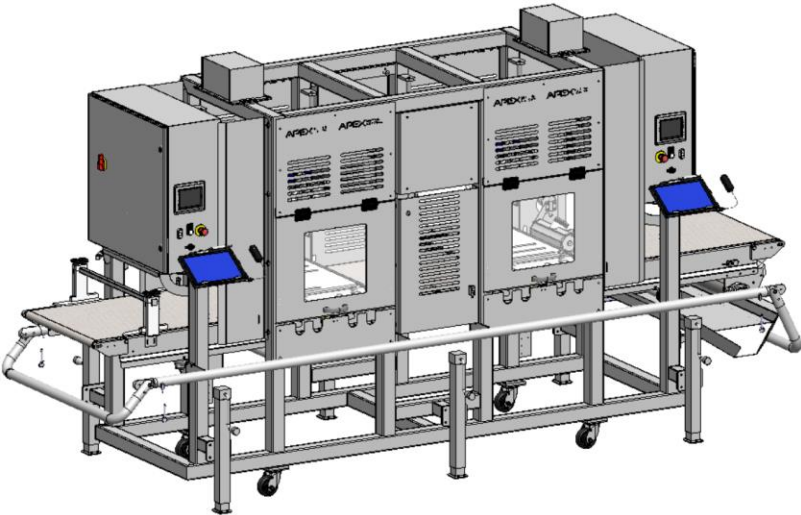


Pick & Place Cell

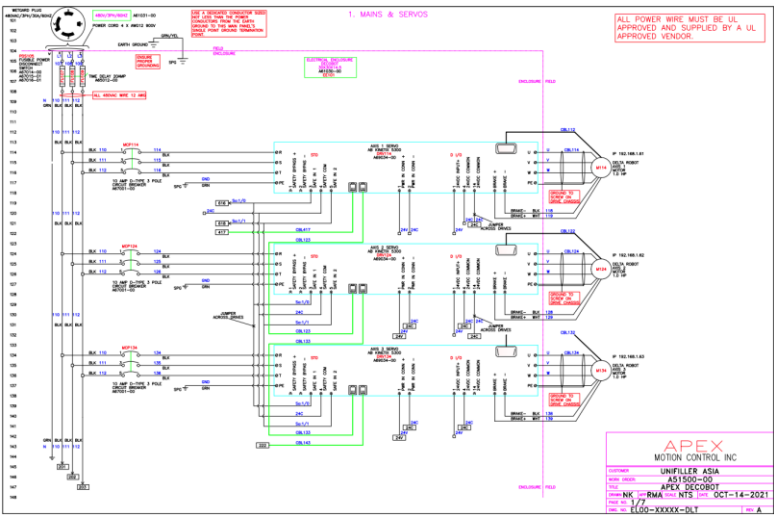


Electrical Panel Layout

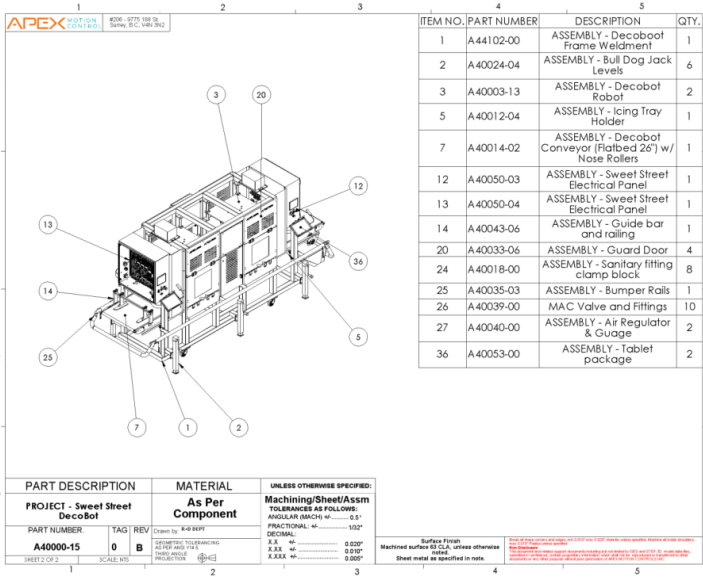
Navraj Kambo | DecoBot



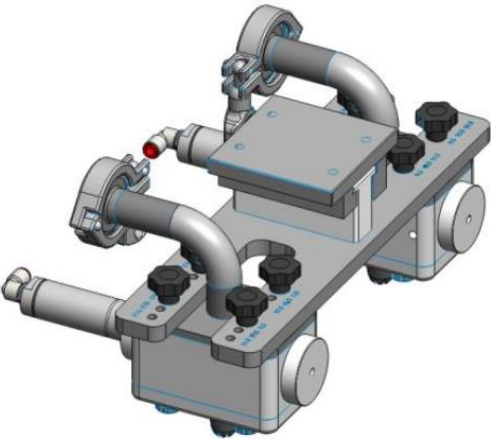
SolidWorks Assembly (Dual DecoBot)



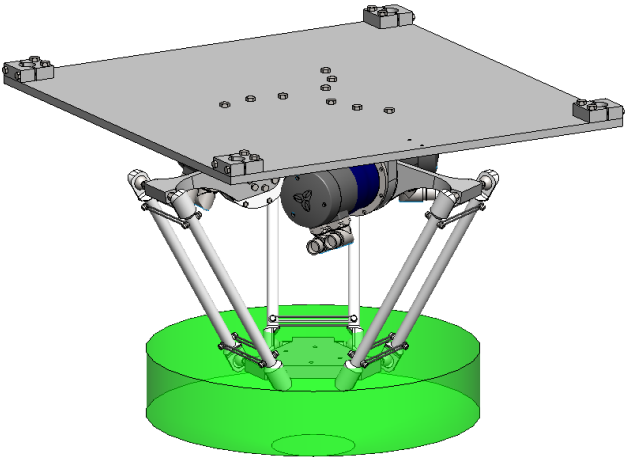
Electrical Design



Machine Drawing



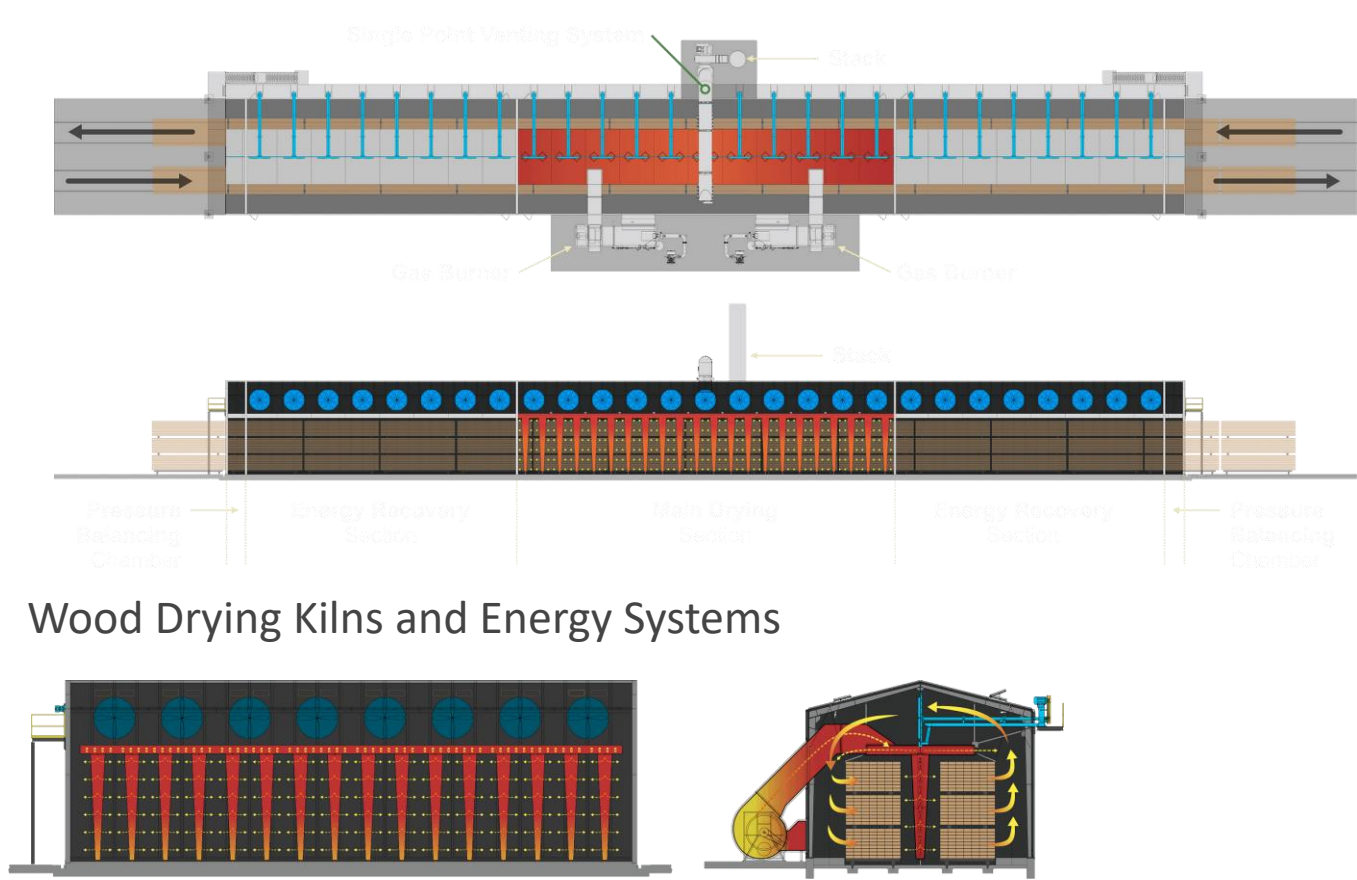
Robot Decorating Nozzle



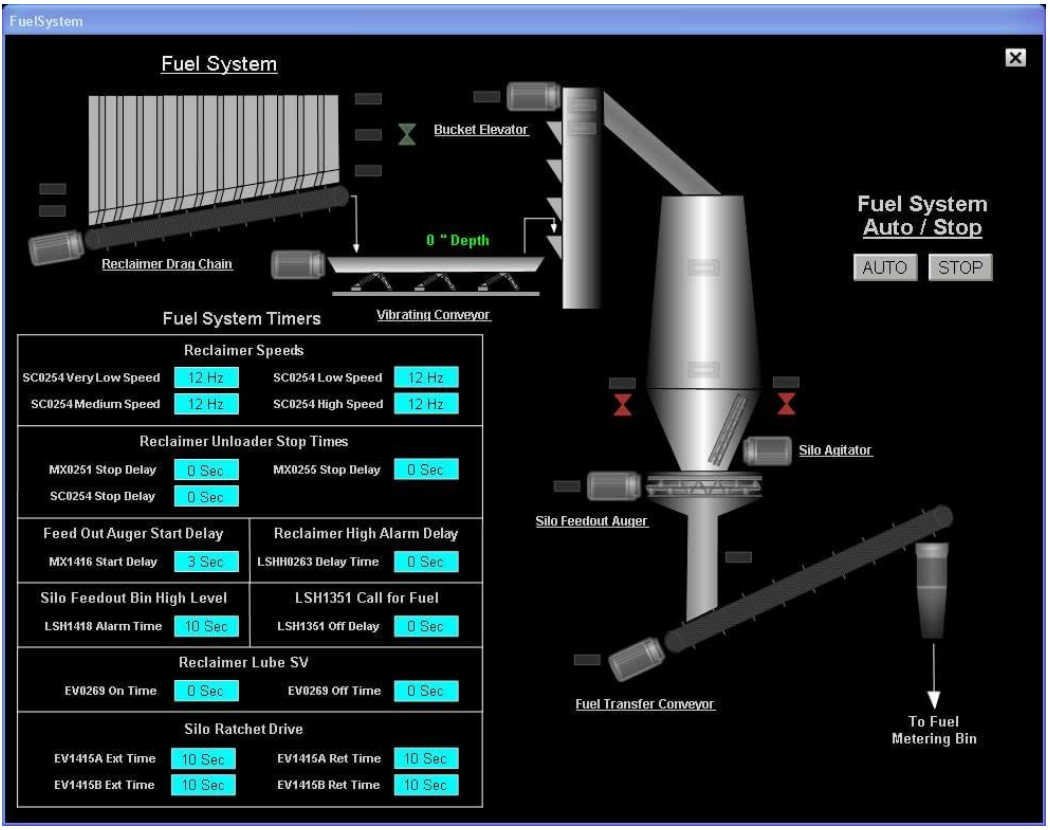
Delta Robot



Actual Robot

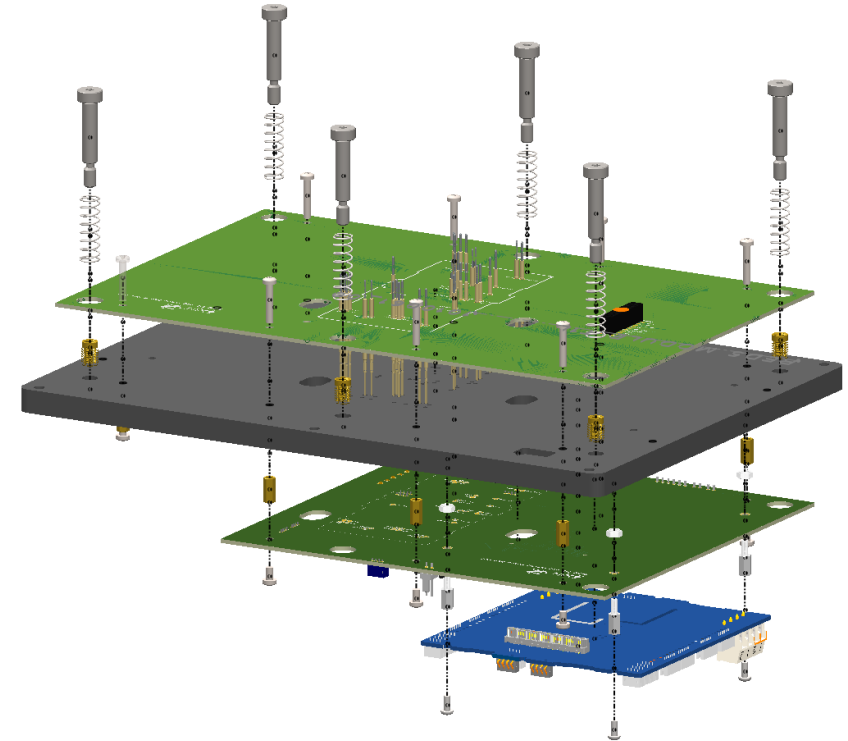
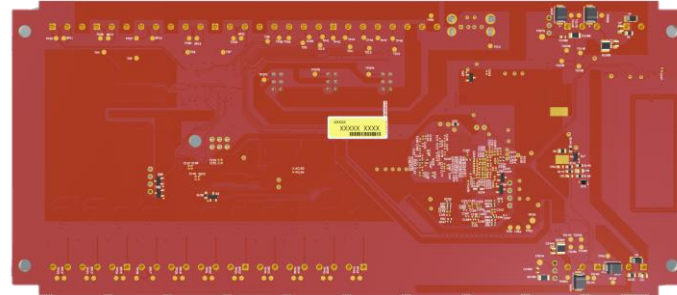
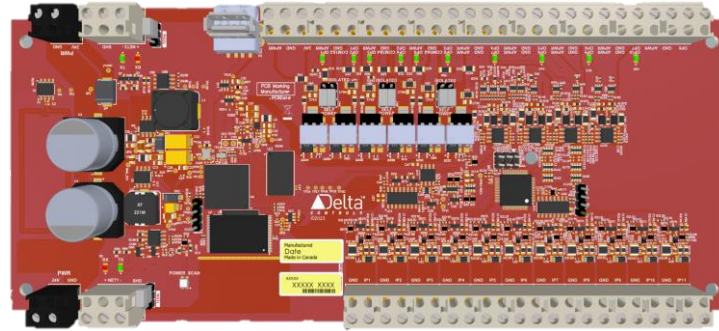
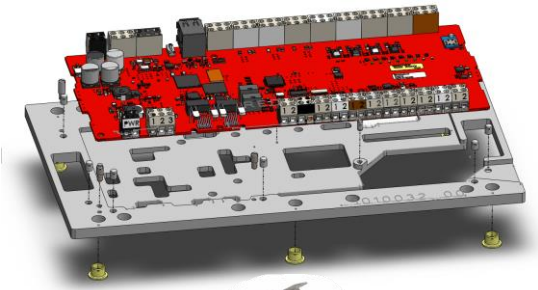


Wood Drying Kilns and Energy Systems

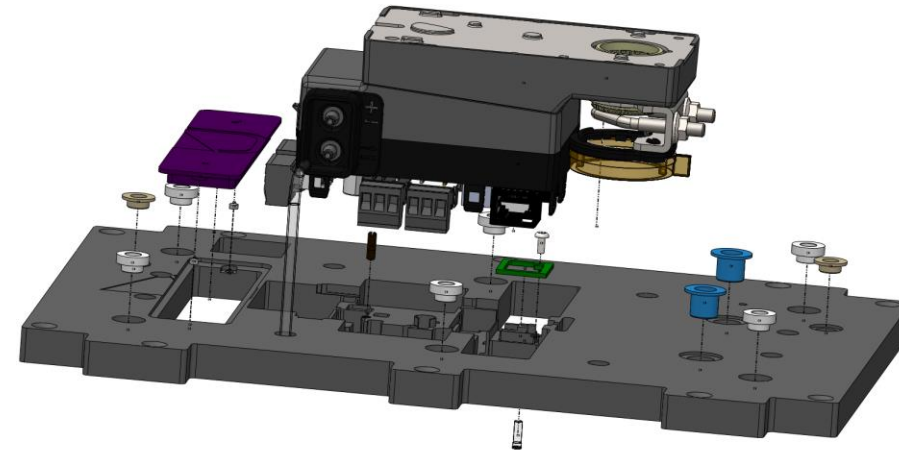
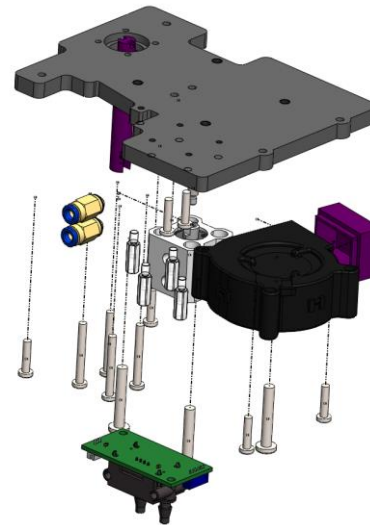
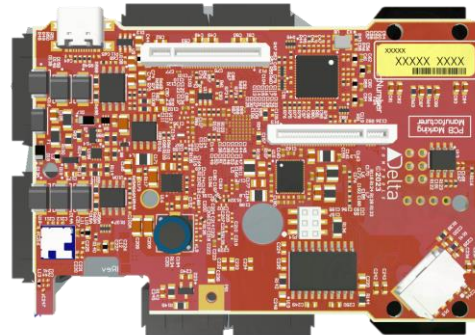
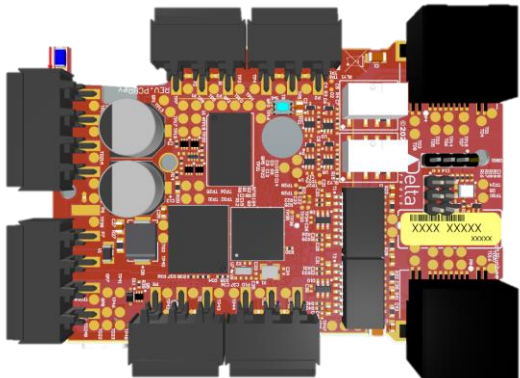




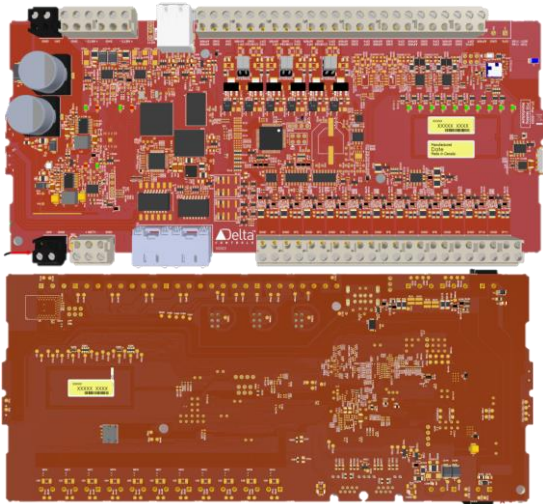
Navraj Kambo | DIBT – TAG Engineer



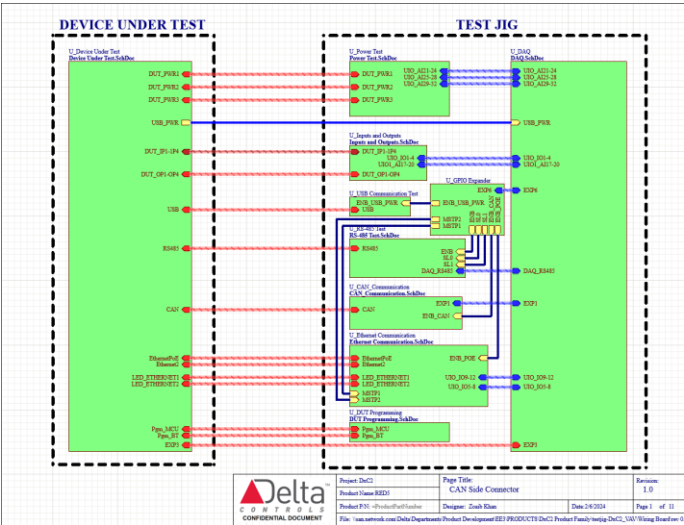
HVAC Controllers and Smart Thermostats (Testing Group)



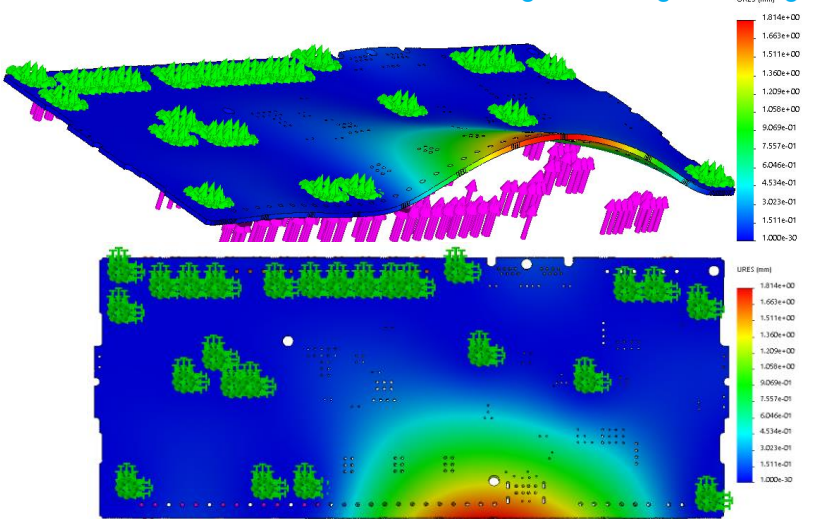
Navraj Kambo | RED 5 Edge (China)



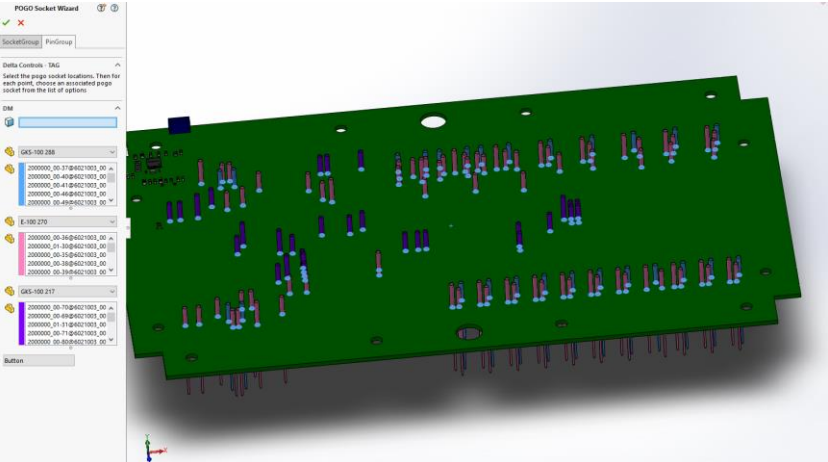
Hardware Review



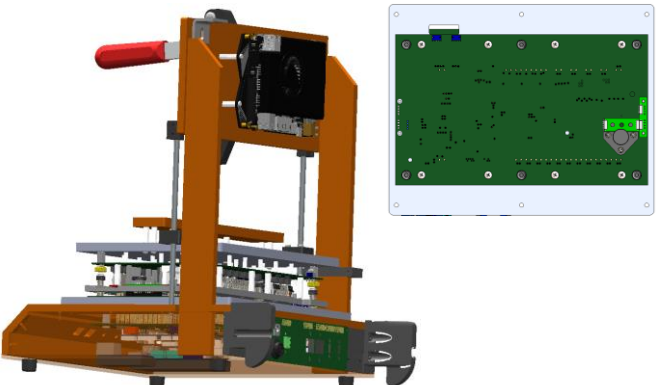
Test Jig Schematic Review



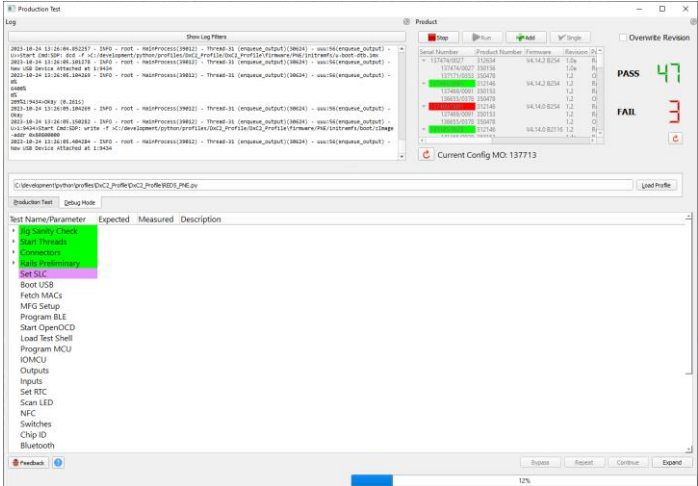
POGO Pin Static Analysis



POGO Pin/Socket Selection

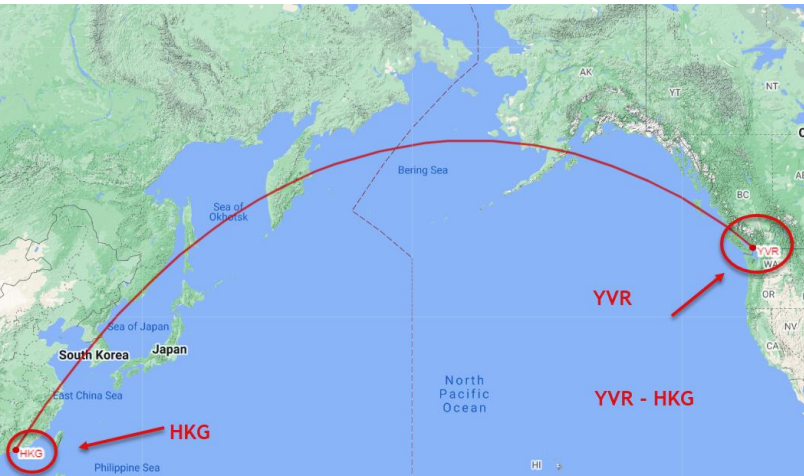


Mechanical Design



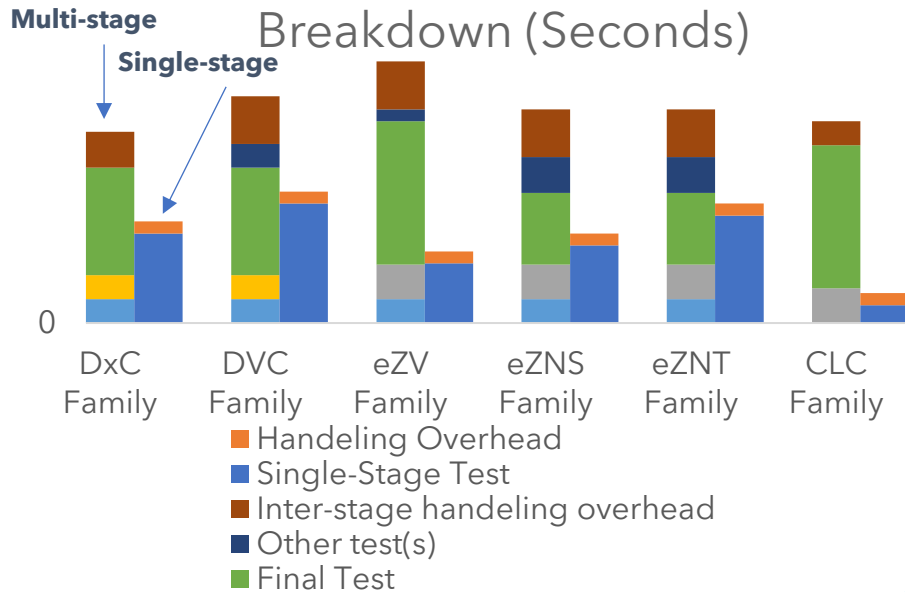
Test Firmware and Framework(UI)

Navraj Kambo | RED 5 Edge (China)

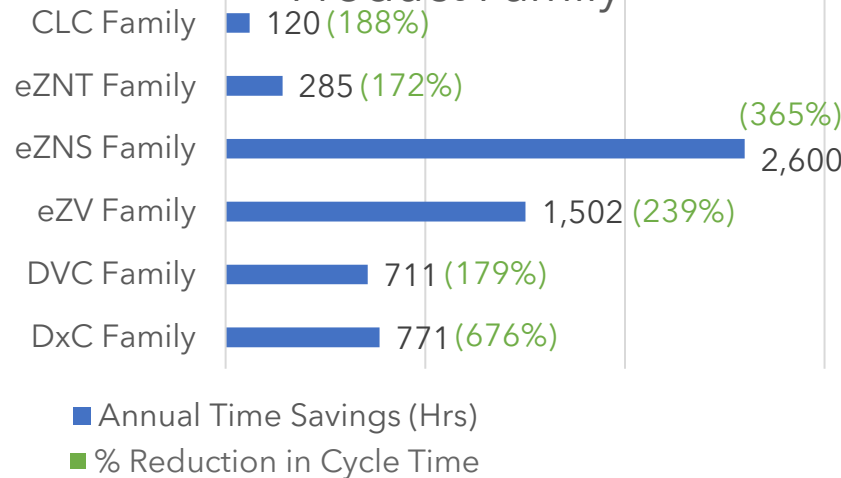


Navraj Kambo | Test and Automation Group

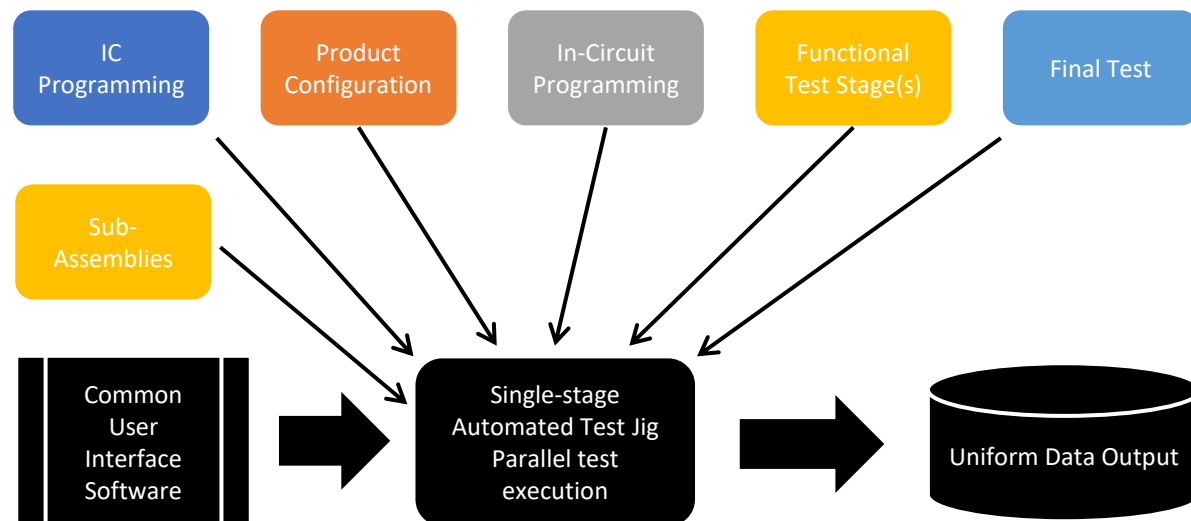
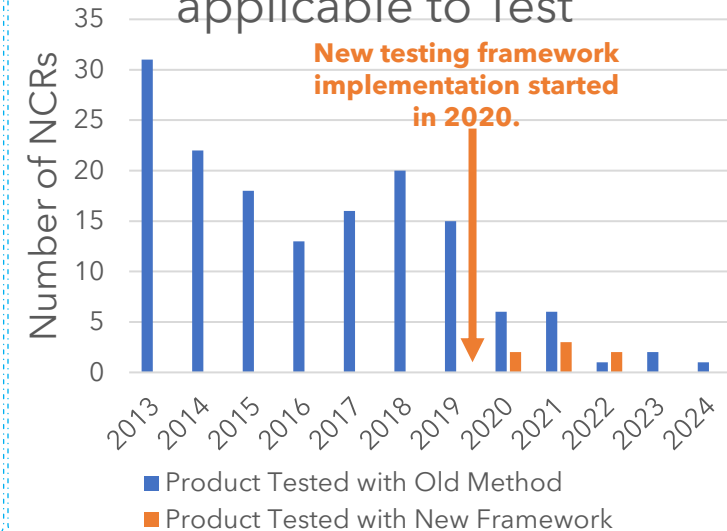
Multi-stage Vs Single-stage Test Breakdown (Seconds)



Annual Time Savings and Cycle Time Reduction by Product Family



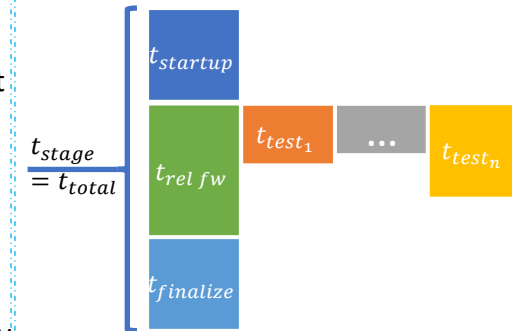
DIBT (Canada) Inc. Non-Conformance Reports applicable to Test



The DIBT (Canada) Inc. Test and Automation team developed a flexible and efficient framework (software, firmware, hardware, and data analysis) that integrates multiple production stages into a single streamlined process.

Key benefits:

- Saves 6,000+ hours annually
- Drastically improves test coverage and data visibility
- Reduces human intervention and standardizes workflows
- Eliminates the need for specialized product training.





Navraj Kambo

Test and Automation Engineer (EIT)

Thank You!

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